

OPENHMI REFERENCE MODULE

TCW-507M

Allwinner T507

English Technical Reference Manual

A balanced platform for connected HMI products that need more graphics and video headroom, dual-display capability and a current 64-bit application processor.

Revision 1.1 | August 2026

Supplier-neutral reference compiled from the supplied module documentation. Confirm the exact hardware revision, pin mapping, BSP status and commercial configuration before design-in.

1. Positioning and selection

This manual summarizes the module from a project-selection perspective. It is intended to help product teams decide whether the platform fits their HMI architecture before requesting detailed carrier-board, software and supply-chain information.

When this module is a good fit

- Industrial and commercial HMI with richer graphics or multimedia
- Linux or Android products requiring 64-bit Cortex-A53 processing
- Dual-display designs using HDMI with LVDS or parallel RGB
- Connected equipment requiring dual Ethernet and multiple USB ports

When to consider another path

- Very cost-constrained products with a simple, single-screen user interface
- Designs requiring native PCIe expansion or an eDP panel interface
- Applications whose lifecycle requires an OS release beyond the documented BSP baseline

Selection rule

Choose the module only after checking display interface, graphics load, boot-time target, operating-system baseline, carrier-board complexity, production lifecycle and the engineering capability available to your team.

2. Core specifications

Processor	Allwinner T507; quad-core Arm Cortex-A53 up to 1.5 GHz
Graphics	Mali-G31 MP2; OpenGL ES 3.2 and Vulkan 1.1 class support
Memory	LPDDR4, typically 2 GB; up to 4 GB by configuration
Storage	eMMC 5.1, typically 8 GB; documented support up to 64 GB
Operating systems	Linux and Android 10 support are documented
Module interface	180 castellated edge pins
Input power	4.8 V to 5.5 V DC; 5 V nominal; carrier design should allow adequate current margin

Configuration note

Memory, storage, wireless and some peripheral options are configuration-dependent. The values above describe documented capability, not a guaranteed stock configuration. Confirm the required combination before quotation or schematic release.

Software baseline

Operating-system support depends on the available BSP, kernel branch, display driver, touch controller, GPU stack and peripheral drivers. Validate the actual software package on target hardware before committing the product schedule.

3. Display, video and touch

Panel outputs	Dual-link LVDS and parallel RGB/CPU interfaces; functions are multiplexed
External video	HDMI output and CVBS output
Resolution	Up to 1080p60 for LVDS/RGB; SoC documentation lists HDMI 2.0a up to 4K60
Multi-display	Two independent paths are documented for HDMI plus LVDS/RGB/CPU combinations
Camera/video input	4-lane MIPI CSI, parallel digital camera and BT.656; up to 8 MP30 is documented
Touch	I2C interface suitable for common capacitive-touch controllers

Display integration checks

- Confirm that the required panel timing, lane count and voltage level are supported by the selected pin-multiplexing mode.
- Check whether dual-display use reduces access to camera, Ethernet, storage or general-purpose I/O functions.
- Validate backlight power, enable and PWM control on the carrier board; these are not supplied by the display data interface alone.
- Confirm the touch-controller interface, reset/interrupt pins, firmware and display bonding stack for the intended panel.
- Run the real UI workload on the target resolution to verify frame rate, memory bandwidth and thermal behavior.

4. Connectivity and I/O

USB	1 x USB 2.0 OTG and 3 x USB 2.0 host
Ethernet	One 10/100/1000 MAC and one 10/100 MAC; dual-port operation is documented
Serial buses	Up to 6 UART, 2 SPI and 6 TWI/I2C at SoC level; routed availability depends on pin multiplexing
Control	Up to 6 PWM, 4 GPADC, 1 LRADC, infrared receiver and RTC support
Audio	Line input, stereo output, I2S/PCM/TDM, S/PDIF and digital microphone options
Wireless	Optional Wi-Fi or combined Wi-Fi/Bluetooth configuration
Other	SD/MMC, transport-stream and smart-card related functions are documented

Pin-multiplexing warning

Interface counts describe the documented processor and module capability. Several functions share pins, clocks or high-speed resources and therefore cannot always be used together. Build a project-specific I/O allocation table before starting the carrier-board schematic.

Request before layout

Obtain the current pin table, carrier-board reference schematic, PCB footprint, power sequence and BSP release note for the exact module revision selected for the project.

5. Mechanical, power and design-in

Module size	60 mm x 40 mm
Overall height	Approximately 3.4 mm
Top-side component height	Approximately 1.8 mm above the module PCB
Carrier keep-out	A carrier-board cutout is required under the bottom-side component area

Power and handling

- Do not hot-plug the module or connect/disconnect it while the carrier board is powered.
- Use a regulated supply with startup margin for display, USB, Ethernet, wireless and storage peak current.
- Follow the documented power sequence, reset timing and signal-voltage domains.
- Provide airflow or heat spreading after measuring the real application workload and enclosure temperature.
- Protect the assembly from condensation, conductive contamination, ESD and mechanical stress.

Project validation checklist

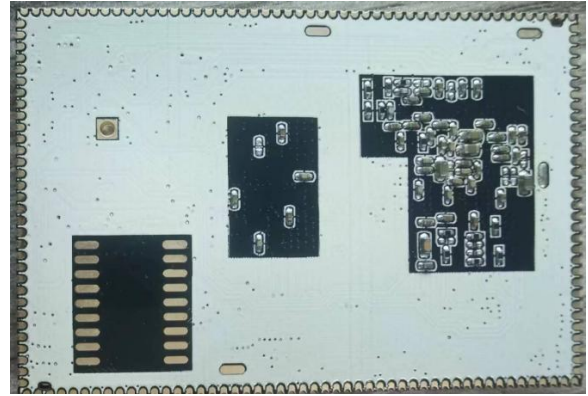
- Exact module revision and production configuration confirmed
- Display and touch panel tested with the intended cable and bonding stack
- All pin conflicts resolved in a carrier-board I/O matrix
- BSP, boot media, update method and recovery path verified
- Thermal, EMC, ESD, power interruption and long-duration operation tested
- Lifecycle, lead time and alternative-component strategy reviewed

6. Hardware views and reference drawings

The photographs below are reproduced from the supplied module documentation. Supplier marks have been redacted while component placement, connectors and edge contacts are left unchanged. Photographs are for identification only and must not be scaled for PCB or enclosure design.



Module top view



Module bottom view

7. Pin definitions and multiplexing reference

The following pages reproduce the original engineering pin tables at high resolution. Signal identifiers and alternate-function names are preserved exactly to reduce transcription risk. Use them as an evaluation reference; the controlled pin table for the selected hardware revision remains the design authority.

English legend for recurring source labels

PIN / GPIO	Physical module pin and associated GPIO bank or signal
Function 1...n	Selectable alternate function through SoC pin multiplexing
Input / Output / I/O	Signal direction: input, output, or bidirectional
Power input / Power output	Supply rail entering or leaving the module
GND	Power and signal ground
NC / Floating	Do not connect unless a controlled revision document states otherwise
Highlighted cells	Default or reserved module functions in the source table; confirm before reassignment

Important

Never route a carrier board from a web manual alone. Request the revision-matched pin spreadsheet, schematic, footprint and power-domain table before schematic sign-off.

Pin definition reference - sheet 1 of 5

A4	MCSI-D1N				
A5	MCSI-CLKP				
A6	MCSI-CLKN				
A7	MCSI-D2P				
A8	MCSI-D2N				
A9	MCSI-D3P				
A10	MCSI-D3N				
A11	LINEOUTL				
A12	LINEOUTR				
A13	USB0-DM				
A14	USB0-DP				
A15	USB1-DM				
A16	USB1-DP				
A17	USB2-DM				
A18	USB2-DP				
A19	USB3-DM				
A20	USB3-DP				
A21	HCEC				
A22	HSDA				
A23	HSCL				
A24	HTXCN				
A25	HTXCP				
A26	HTX0N				
A27	HTX0P				
A28	HTX1N				
A29	HTX1P				
A30	HTX2N				
A31	HTX2P				
A32	HHPD				
A33	PD10/EINT10	LVDS1-D0P	LCD-D10	TS-D6	
A34	PD11/EINT11	LVDS1-D0N	LCD-D11	TS-D7	
A35	PD12/EINT12	LVDS1-D1P	LCD-D12	SMC-VPPEN	
A36	PD13/EINT13	LVDS1-D1N	LCD-D13	SMC-VPPPP	
A37	PD14/EINT14	LVDS1-D2P	LCD-D14	SMC-PWREN	
A38	PD15/EINT15	LVDS1-D2N	LCD-D15	SMC-CLK	
A39	PD16/EINT16	LVDS1-CLKP	LCD-D16	SMC-DATA	
A40	PD17/EINT17	LVDS1-CLKN	LCD-D17	SMC-RST	
A41	PD18/EINT18	LVDS1-D3P	LCD-D18	SMC-DET	
A42	PD19/EINT19	LVDS1-D3N	LCD-D19		
A43	PD0/EINT0	LVDS0-D0P	LCD-D0	TS-CLK	
A44	PD1/EINT1	LVDS0-D0N	LCD-D1	TS-ERR	
A45	PD2/EINT2	LVDS0-D1P	LCD-D2	TS-SYNC	
A46	PD3/EINT3	LVDS0-D1N	LCD-D3	TS-DVLD	
A47	PD4/EINT4	LVDS0-D2P	LCD-D4	TS-D0	
A48	PD5/EINT5	LVDS0-D2N	LCD-D5	TS-D1	
A49	PD6/EINT6	LVDS0-CLKP	LCD-D6	TS-D2	

Original engineering table, source page 9. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 2 of 5

A50	PD7/EINT7	LVDS0-CLKN	LCD-D7	TS-D3		
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C 端口如下:

PIN	GPIO 功能	功能 1	功能 2	功能 3	功能 4	功能 5
C1	PD8/EINT8	LVDS0-D3P	LCD-D8	TS-D4		
C2	PD9/EINT9	LVDS0-D3N	LCD-D9	TS-D5		
C3	PD20/EINT20	LCD-D20				
C4	PD21/EINT21	LCD-D21				
C5	PD22/EINT22	LCD-D22				
C6	PD23/EINT23	LCD-D23				
C7	PD28/EINT28	LCD-PWM				
C8	PD25/EINT25	LCD-DE				
C9	PD27/EINT27	LCD-VSYNC				
C10	PD26/EINT26	LCD-HSYNC				
C11	PD24/EINT24	LCD-CLK				
C12	PA12/EINT12	RMII-RESET	WATCHDONG-SIG			
C13	PA7/EINT7	RMII-TXEN	I2S0-BCK	AC-SYNC		
C14	PA6/EINT6	RMII-TXCK	I2S0-MCLK	AC-MCLK		
C15	PA9/EINT9	RMII-MDIO	I2S0-DIN	AC-ADCR		
C16	PA0/EINT0	RMII-RXD1	TWI0-SCK			
C17	PA8/EINT8	RMII-MDC	I2S0-LRCK	AC-ADCL		
C18	PA10/EINT10	TWI3-SCK				
C19	PA4/EINT4	RMII-TXD1	AC-ADCY	STANDBY-STATE		
C20	PA5/EINT5	RMII-TXD0	I2S0-DOUT0	AC-ADCX		
C21	PA1/EINT1	RMII-RXD0	TWI0-SDA			
C22	PA2/EINT2	RMII-CRS-DV	TWI1-SCK			
C23	PA3/EINT3	RMII-RXER	TWI1-SDA			
C24	PA11/EINT11	TWI3-SDA				
C25	PH4/EINT4	SPDIF-OUT	TWI3-SCK			
C26	PH1/EINT1	UART0-RX	PWM4	TWI1-SDA		
C27	PH0/EINT0	UART0-TX	PWM3	TWI1-SCK		
C28	PH5/EINT5	SPI1-CS0	I2S3-MCLK	UART2-TX	TWI3-SDA	
C29	PH6/EINT6	SPI1-CLK	I2S3-BCK	UART2-RX	TWI4-SCK	
C30	PH3/EINT3	UART5-RX	SPDIF-IN	PWM1	TWI2-SDA	
C31	PH2/EINT2	UART5-TX	SPDIF-CLK	PWM2	TWI2-SCK	
C32	PH10/EINT10	IR-RX	TCON-TRIG1			

B 端口如下:

PIN	GPIO 功能	功能 1	功能 2	功能 3	功能 4	功能 5
B1	PF0/EINT0	SDC0-D1	JTAG-MS			
B2	PF1/EINT1	SDC0-D0	JTAG-DI			

Original engineering table, source page 10. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 3 of 5

B3	PI14/EINT14	RGMII-MDC	UART4-RX	TS0-D5	PWM4	
B4	PI13/EINT13	RGMII-CLKIN-125M	RMII-NULL	UART4-TX	TS0-D4	PWM3
B5	PI12/EINT12	RGMII-TXCTL	RMII-TXEN	UART3-CTS	TS0-D3	PWM2
B6	PI11/EINT11	RGMII-TXCK	RMII-TXXCK	UART3-RTS	TS0-D2	PWM1
B7	PI10/EINT10	RGMII-TXD0	RMII-TXD0	UART3-RX	TS0-D1	TWI2-SDA
B8	PI9/EINT9	RGMII-TXD1	RMII-TXD1	UART3-TX	TS0-D0	TWI2-SCK
B9	PI8/EINT8	RGMII-TXD2	RMII-NULL	UART2-CTS	TS0-DVLD	TWI1-SDA
B10	PI7/EINT7	RGMII-TXD3	RMII-NULL	UART2-RTS	TS0-SYNC	TWI1-SCK
B11	PI6/EINT6	PHYRSTB	RGMII-NULL/ RMII-RXER	UART2-RX	TS0-ERR	TWI0-SDA
B12	PI5/EINT5	RGMII-RXCTL	RMII-CRS-DV	UART2-TX	TS0-CLK	TWI0-SCK
B13	PI4/EINT4	RGMII-RXCK	DMIC-DATA3	RMII-NULL	I2S0-DIN0	I2S0-DOUT1
B14	PI3/EINT3	RGMII-RXD0	DMIC-DATA2	RMII-RXD0	I2S0-DOUT0	I2S0-DIN1
B15	PI2/EINT2	RGMII-RXD1	DMIC-DATA1	RMII-RXD1	I2S0-LRCK	HDMI-CEC
B16	PI1/EINT1	RGMII-RXD2	DMIC-DATA0	RMII-NULL	I2S0-BCLK	HDMI-SDA
B17	PI0/EINT0	RGMII-RXD3	DMIC-SCK	RMII-NULL	I2S0-MCLK	HDMI-SCL
B18	PE22/EINT23	NCSI0-FSIN0	TCON-TRIG0			
B19	PE4/EINT5	NCSI0-D0				
B20	PE5/EINT6	NCSI0-D1				
B21	PE6/EINT7	NCSI0-D2				
B22	PE7/EINT8	NCSI0-D3				
B23	PE8/EINT9	NCSI0-D4				
B24	PE9/EINT10	NCSI0-D5				
B25	PE10/EINT11	NCSI0-D6				
B26	PE11/EINT12	NCSI0-D7				
B27	PE12/EINT13	NCSI0-D8				
B28	PE13/EINT14	NCSI0-D9				
B29	PE14/EINT15	NCSI0-D10				
B30	PE15/EINT16	NCSI0-D11				
B31	PE16/EINT17	NCSI0-D12				
B32	PE17/EINT18	NCSI0-D13				
B33	PE18/EINT19	NCSI0-D14				
B34	PE19/EINT20	NCSI0-D15				
B35	PE0/EINT1	NCSI0-PCLK				
B36	PE21/EINT22	NCSI0-SDA	TWI2-SDA			
B37	PE20/EINT21	NCSI0-SCK	TWI2-SCK			
B38	PE3/EINT4	NCSI0-VSYNC				
B39	PE2/EINT3	NCSI0-HSYNC				
B40	PE1/EINT2	NCSI0-MCLK				
B41	GND					
B42	PG19/EINT19	MCSI-MCLK	PWM1			
B43	PG18/EINT18	MCSI-SDA	TWI3-SDA	UART2-CTS		
B44	PG17/EINT17	MCSI-SCK	TWI3-SCK	UART2-RTS		
B45	PG16/EINT16	BT-WAKE-AP	TWI4-SDA	UART2-RX		
B46	PG15/EINT15	WL-WAKE-AP	TWI4-SCK	UART2-TX		
B47	IRQ	4G-WAKEUP-S				

Original engineering table, source page 11. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 4 of 5

		OC				
B48	PH9/EINT9	SPI1-CS1	I2S3-DIN0	I2S3-DOUT1		
B49	PH8/EINT8	SPI1-MISO	I2S3-DOUT0	I2S3-DIN1	UART2-CTS	UART2-CTS
B50	PH7/EINT7	SPI1-MOSI	I2S3-LRCK	UART2-RTS	TWI4-SDA	

D 端口如下:

PIN	GPIO 功能	功能 1	功能 2	功能 3	功能 4	功能 5
D1	LRADC	KEY 按键				
D2	GPADC1	ADC 输入				
D3	GPADC2	ADC 输入				
D4	GPADC3	ADC 输入				
D5	TVOUT	CVBS 信号				
D6	GND	电源地				
D7	CLDO1	0.7~3.3V 可调输出 200mA				
D8	CLDO4	0.7~3.3V 可调输出 200mA				
D9	POWER-ON	电源按键				
D10	RESET	复位按键				
D11	BLDO2	0.7~3.3V 可调输出 500mA (核芯板提供给外围应用, 开, 关电源可控制)				
D12	BLDO3	0.7~3.3V 可调输出 300mA (核芯板提供给外围应用, 开, 关电源可控制)				
D13	BLDO5	0.7~3.3V 可调输出 600mA (核芯板提供给外围应用, 开, 关电源可控制)				
D14	DCDC1	3.3V 电源输出 1A (核芯板提供给外围应用, 开, 关电源不可控制)				
D15	DCDC6	0.5~3.4V 可调输出 2.0A (核芯板提供给外围应用, 开, 关电源可控制)				
D16	DCDC6					
D17	DCIN5V	DC5V/3A 输入 (核芯板供电电源)				
D18	DCIN5V					
D19	GND	电源地输入				
D20	GND					
D21	FEL	BOOT 键输入				
D22	LINEINL	音频输入左声道				
D23	LINEINR	音频输入右声道				
D24	PI15/EINT15	RGMII-MDIO	UART-RTS	TS0-D6	CLK-FANOUT0	
D25	PI16/EINT16	EPHY-25M	UART-CTS	TS0-D7	CLK-FANOUT1	
D26	PF5/EINT5	SDC0-D2	JTAG-CK			
D27	PF6/EINT6	SDC0-DET				
D28	PF3/EINT3	SDC0-CMD	JTAG-DO			
D29	ANT	WIFI/BT 天线				
D30	GND	电源地				
D31	PF2/EINT2	SDC0-CLK	UART0-TX			
D32	PF4/EINT4	SDC0-D3	UART0-RX			

E 端口如下:

PIN	GPIO 功能	功能 1	功能 2	功能 3	功能 4	功能 5
E1	FMINL	FM 左声道输入				

Original engineering table, source page 12. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 5 of 5

E2	FMINR	FM 右声道输入				
E3	GND	电源地				
E4	PG3/EINT3	WL-SDIO-D1				
E5	PG2/EINT2	WL-SDIO-D0				
E6	PG0/EINT0	WL-SDIO-CLK				
E7	PG11/EINT11	BT-PCM-CLK	I2S2-BCK	BIST-RESULT0	AC-SYNC	
E8	PG10/EINT10	AP-CK32KO	I2S2-MCLK	AC-MCLK		
E9	PG14/EINT14	BT-PCM-DOUT	I2S2-DIN0	I2S2-DOUT1	BIST-RESULT3	AC-ADCX
E10	PG12/EINT12	BT-PCM-SYNC	I2S2-LRCK	BIST-RESULT1	AC-ADCL	
E11	PG13/EINT13	BT-PCM-DIN	I2S2-DOUT0	I2S2-DIN1	BIST-RESULT2	AC-ADCR
E12	PG6/EINT6	BT-UART-TX	JTAG-MS			
E13	PG7/EINT7	BT-UART-RX	JTAG-CK			
E14	PG9/EINT9	BT-UART-CTS	JTAG-DI	AC-ADCY		
E15	PG8/EINT8	BT-UART-RTS	JTAG-DO	PLL-LOCK-DBG		
E16	PG1/EINT1	WL-SDIO-CMD				
E17	PG5/EINT5	WL-SDIO-D3				
E18	PG4/EINT4	WL-SDIO-D2				
F19	GND	电源地				

Original engineering table, source page 13. Signal names are preserved; see the English legend on the preceding page.

8. Document scope and next step

This is an English selection and integration reference. It intentionally excludes supplier branding, contact details, service claims and unverified commercial statements. Original photos and engineering-table facsimiles are included where they help identification or pin evaluation, with supplier marks redacted. This manual is not a replacement for the controlled pinout, schematic, mechanical drawing, processor datasheet or BSP documentation for a released production configuration.

Recommended next step

Share the product application, display size and interface, target resolution, operating system, required I/O, annual volume range, environmental conditions and development capability. The module can then be matched against the real project requirements and compared with chip-level or more integrated HMI paths.

Discuss a similar project

Visit openhmi.network to request relevant platform, display, touch and engineering expertise for your application.

Document control

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